



INFORME EXPERT – INFORME EXPERTO EXTERNO – EXTERNAL REFEREE REPORT

Nom de l'expert / Nombre del experto / Name of the referee:

Categoria / Categoría / Category:

Departament, Universitat a què pertany / Departamento, Universidad a la que pertenece /
Department, University to which s/he belongs:

Títol de la tesi oficial / Título de la tesis oficial / Title of the official thesis:

Hardware-Software Co-Design for Accelerating Sparse Tensor Algebra

Nom del doctorand que presenta la tesi:

MARCO SIRACUSA

Especifiqueu les raons que avalen la qualitat de la tesi per a la seva defensa pública:

Objectius assolits; Originalitat del treball; Metodologia emprada; Hipòtesis contrastades; Qualitat del manuscrit

Especificar los motivos que avalan la calidad de la tesis mencionada para su defensa pública:

Objetivos logrados; Originalidad del trabajo; Metodología usada; Hipótesis contrastadas; Calidad del manuscrito

Specify reasons endorsing the quality of the above-mentioned thesis for its public reading:

Objectives achieved; Originality of the work; Methodology used; Hypotheses tested, quality of the manuscript

The thesis moves from the observations that sparse computations are increasingly important in scientific computing and machine learning, but today's CPUs, GPUs, supercomputers, and datacenters are mainly designed for dense, regular computation. Because of this mismatch, sparse workloads often fail to use available hardware efficiently.

The key ideas are:

1. Sparse workloads are inefficient on traditional hardware: Sparse tensor operations require irregular memory access, tensor traversal, and coordinate merging. Conventional compute cores are not well suited to these tasks, so performance is often far below the hardware's potential.
2. Existing sparse-computing solutions are limited: Prior approaches often focus on narrow types of sparse tensor operations, underuse modern vector units, or require users to manage too much low-level complexity.

The thesis achieves the objective of defining and demonstrating a hardware–software co-design approach for designing systems specifically for general sparse tensor algebra, combining new hardware support with compiler automation. A Decoupled Access–Execute architecture is introduced: The proposed architecture separates sparse data access and traversal from numerical computation. A specialized Tensor Marshaling Unit handles irregular tensor traversal and coordinate merging, while the main core performs computation.

The key hypothesis tested is that the approach improves performance significantly with respect to traditional architectures: By offloading sparse marshaling work to the TMU, the architecture can outperform traditional CPUs and GPUs by up to an order of magnitude.

The methodology used in the thesis pushes a hardware–software codesign approach. In fact, the thesis also presents Ember, an MLIR-based compiler that automatically lowers sparse machine learning operations to DAE/TMU code. This allows integration with frameworks such as PyTorch and TensorFlow without forcing users to manually program the specialized architecture.

Overall, the thesis presents a way to make general sparse tensor computation both faster and more programmable by combining specialized sparse hardware with compiler support.

The quality of the manuscript is high, in terms of writing and in terms of content and results.

Valoració absoluta i/o ponderada de la tesi en relació amb altres treballs d'investigació:

Valoración absoluta y/o ponderada de la tesis presentada en comparación con otros trabajos de investigación:

Absolute and/or relative assessment of the thesis in comparison with other works of research:

Aquesta tesi està entre les millors / Esta tesis está entre las mejores / This thesis is among the top:
Excel·lent / Excelente / Outstanding

Considera que la tesi esmentada és apta per al tràmit de lectura i la defensa pública?

¿Considera que la tesis anteriormente mencionada es apta para su lectura y defensa pública?

In consideration of all the above, is the thesis judged to be suitable for public reading?



Sí / Yes



Sí amb correccions majors / Sí con correcciones mayores / Yes with major corrections



Sí amb correccions menors / Sí con correcciones menores / Yes with minor corrections



No

Observacions / Observaciones / Observations:

I liked particularly the hardware-software codesign angle, not only looking into hardware and architecture, but also emphasizing the code generation and optimization.